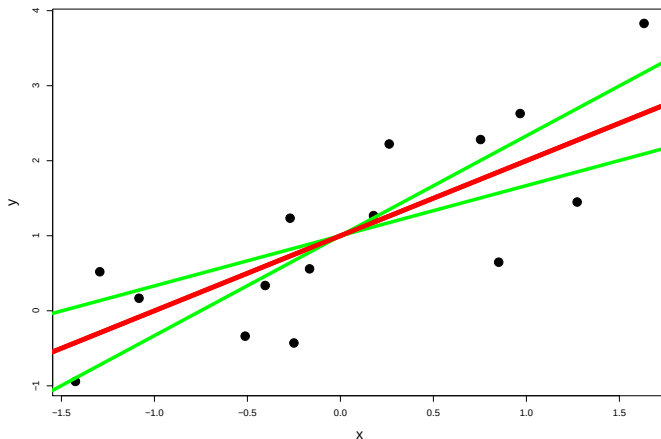


Machine Learning

Volker Roth

Department of Mathematics & Computer Science
University of Basel

Chapter 3: Regression



Least-squares fit (red) and two lines with slopes according to upper (lower) 95% confidence limit (green).

Common continuous distributions: Multivariate Normal

- The multivariate normal distribution of a k -dimensional random vector $\mathbf{X} = (X_1, \dots, X_k)^t$ can be written as: $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, with k -dimensional **mean vector**

$$\boldsymbol{\mu} = E[\mathbf{X}] = [E[X_1], E[X_2], \dots, E[X_k]]^t$$

and $k \times k$ **covariance matrix**

$$\Sigma =: E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^t] = [\text{Cov}[X_i, X_j]; 1 \leq i, j \leq k],$$

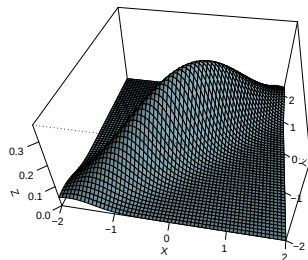
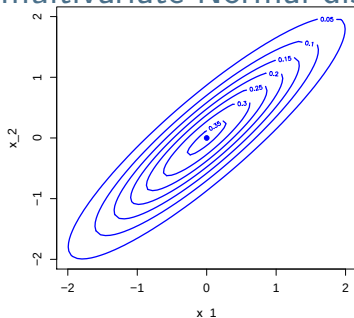
where

$$\text{Cov}[X_i, X_j] = E[(X_i - \mu_i)(X_j - \mu_j)].$$

- The inverse of the covariance matrix is called **precision matrix**
- The pdf of the multivariate normal distribution is

$$p(x_1, \dots, x_k | \boldsymbol{\mu}, \Sigma) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^t \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

The multivariate Normal distribution



Affine transformations:

If $\mathbf{Y} = \mathbf{c} + B\mathbf{X}$ is an affine transformation of $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, then $\mathbf{Y} \sim \mathcal{N}(\mathbf{c} + B\boldsymbol{\mu}, B\Sigma B^t)$. Why?

$$\boldsymbol{\mu}_Y = E[\mathbf{c} + B\mathbf{X}] = \mathbf{c} + BE[\mathbf{X}] = \mathbf{c} + B\boldsymbol{\mu}$$

$$\begin{aligned}\Sigma_Y &= E[(\mathbf{c} + B\mathbf{X} - \mathbf{c} + B\boldsymbol{\mu})(\mathbf{c} + B\mathbf{X} - \mathbf{c} + B\boldsymbol{\mu})^t] \\ &= E[B(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^t B^t] = B\Sigma B^t\end{aligned}$$

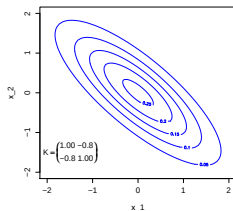
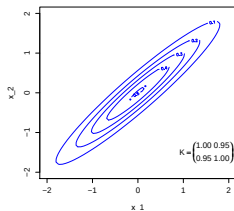
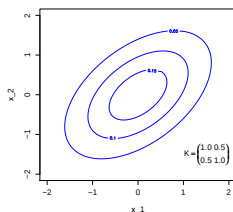
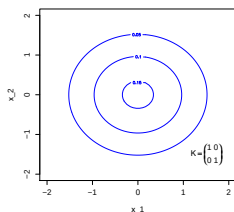
The multivariate normal distribution

$$\text{2D Gaussian: } p(\mathbf{x}|\boldsymbol{\mu} = \mathbf{0}, \Sigma) = \frac{1}{\sqrt{2\pi|\Sigma|}} \exp\left(-\frac{1}{2}\mathbf{x}^t \Sigma^{-1} \mathbf{x}\right)$$

Covariance

(also written “co-variance”) is a measure of how much **two random variables vary together**:

- **positive**: positive linear coherence,
- **negative**: negative linear coherence,
- **0**: no linear coherence.



Regression basics

- In regression we assume that a response variable $y \in \mathbb{R}$ is a noisy function of the input variable $\mathbf{x} \in \mathbb{R}^d$.

$$y = f(\mathbf{x}) + \eta.$$

- We often assume that f is linear, $f(\mathbf{x}) = \mathbf{w}^t \mathbf{x}$, and that η has a zero-mean Gaussian distribution with constant variance, $\eta \sim N(0, \sigma^2)$.
- This can equivalently be written as

$$p(y|\mathbf{x}) = N(\mu(\mathbf{x}), \sigma^2), \text{ with } \mu(\mathbf{x}) = \mathbf{w}^t \mathbf{x}.$$

- In one dimension: $\mu(\mathbf{x}) = w_0 + w_1 x$ and $\mathbf{x} = (1, x)$.
 w_0 is the **intercept** or bias term and w_1 is the **slope**.
- If $w_1 > 0$, we expect the output to increase as the input increases.

Least Squares and Maximum Likelihood

- Fit n data points $(\mathbf{x}_i, y_i)$ to a model that has $d + 1$ parameters $w_j, j = 0, \dots, d$.
- Notation: $\mathbf{x} \leftarrow (1, \mathbf{x}) \rightsquigarrow w_0$ is the intercept.
- Frequentist view: $\mathbf{w}$ is an unknown parameter vector, not a RV.
- We assume that the n observations are **iid**.
- **Linear model:** $y_i = \mathbf{w}^t \mathbf{x}_i + \eta_i, \quad \eta_i \sim N(0, \sigma^2)$.
Observed y_i generated from a normal distribution centered at $\mathbf{w}^t \mathbf{x}_i$.
- Model predicts linear relationship between **conditional expectation** of **observations** y_i and **inputs** $\mathbf{x}_i$:

$$E[y_i | \mathbf{x}_i] = w_0 + w_1 x_{i1} + \dots + w_d x_{id} = \mathbf{w}^t \mathbf{x}_i = f(\mathbf{x}_i; \mathbf{w}).$$

Note: the expectation operator is linear and $E[\eta_i] = 0$.

Regression function = conditional expectation.

LS and Maximum Likelihood

- **Likelihood function:** conditional probability of all observed y_i given their explanation, treated as a function of the model parameters $\mathbf{w}$:

$$L(\mathbf{w}) \propto \prod_i \exp \left[-\frac{1}{2\sigma^2} (y_i - \mathbf{w}^t \mathbf{x}_i)^2 \right]$$

- **Maximizing L = finding model that best explains observations:**

$$\begin{aligned} \hat{\mathbf{w}} &= \arg \max_{\mathbf{w}} L(\mathbf{w}) = \arg \min_{\mathbf{w}} [-L(\mathbf{w})] = \arg \min_{\mathbf{w}} [-\log(L(\mathbf{w}))] \\ &= \arg \min_{\mathbf{w}} \sum_i (y_i - \mathbf{w}^t \mathbf{x}_i)^2 \end{aligned}$$

Least-squares fit = ML estimate under Gaussian error model.

- $\hat{\mathbf{w}}_{MLE}$ minimizes the **residual sum of squares**

$$RSS(\mathbf{w}) = \sum_{i=1}^n r_i^2 = \sum_{i=1}^n [y_i - f(\mathbf{x}_i; \mathbf{w})]^2 = \|\mathbf{y} - X\mathbf{w}\|^2.$$

LS and Maximum Likelihood

- Finding the optimal weights:

$$\frac{\partial \text{RSS}(\mathbf{w})}{\partial \mathbf{w}} = \frac{\partial}{\partial \mathbf{w}} [\mathbf{y}^t \mathbf{y} - 2\mathbf{y}^t X \mathbf{w} + \mathbf{w}^t X^t X \mathbf{w}] \stackrel{!}{=} \mathbf{0}$$

- Using the following results from matrix calculus,

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{y}^t \mathbf{x} = \mathbf{y}$$

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{x}^t M \mathbf{x} = 2M\mathbf{x}, \text{ if } M \text{ is symmetric,}$$

we finally arrive at

$$\hat{\mathbf{w}} = (X^t X)^{-1} X^t \mathbf{y}.$$

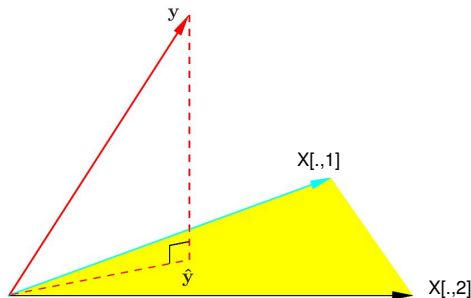
Least squares regression: Geometry

The **residual** is $r = y - Xw$. **Gradient at $w = \hat{w}$ vanishes.**

$$\hat{w} = (X^t X)^{-1} X^t y \Rightarrow X^t (y - X \hat{w}) = X^t r = 0.$$

It follows that $\sum_{i=1}^n X_{ij} r_i = 0$, $\forall j = 0, 1, \dots, d$.

$\leadsto$ **Residual is orthogonal to every input dimension $X_{\bullet j}$.**



Adapted from Fig. 3.2 in (Hastie, Tibshirani, Friedman: The Elements of Statistical Learning Theory. Springer)

Frequentist confidence limits

- **Recall:** $y_i = f(\mathbf{x}_i; \mathbf{w}) + \eta_i$, with independent Gaussian noise.
- In matrix-vector form: $\mathbf{y} = X\mathbf{w} + \boldsymbol{\eta}$, with $\boldsymbol{\eta} \sim N(\mathbf{0}, \sigma^2 I_n)$.

$$\begin{aligned}\hat{\mathbf{w}} &= (X^t X)^{-1} X^t \mathbf{y} \\ &= (X^t X)^{-1} X^t X \mathbf{w} + (X^t X)^{-1} X^t \boldsymbol{\eta} \\ &= \mathbf{w} + (X^t X)^{-1} X^t \boldsymbol{\eta}\end{aligned}$$

$$\Rightarrow \quad \hat{\mathbf{w}} - \mathbf{w} = (X^t X)^{-1} X^t \boldsymbol{\eta} =: A\boldsymbol{\eta}$$

- **Linear** functions of normals are normal:

$$\boldsymbol{\eta} \sim N(\mathbf{0}, \sigma^2 I_n) \Rightarrow A\boldsymbol{\eta} \sim N(\mathbf{0}, \sigma^2 A A^t).$$

$$\text{Here: } A = (X^t X)^{-1} X^t \Rightarrow A A^t = (X^t X)^{-1}$$

- Conditioned on X and σ^2 :

$$\hat{\mathbf{w}} - \mathbf{w} | X, \sigma^2 \sim N\left(\mathbf{0}, \sigma^2 (X^t X)^{-1}\right).$$

Frequentist confidence limits

- Distribution completely specified $\rightsquigarrow$ **confidence limits:**

For k -th component: $\hat{w}_k - w_k \sim N(0, \sigma^2 S^{kk})$,
where S^{kk} denotes the k -th diagonal element of $(X^t X)^{-1}$.

- Thus, z_k is standard normal

$$z_k := (w_k - \hat{w}_k) / \sqrt{\sigma^2 S^{kk}} \sim N(0, 1)$$

- CDF:

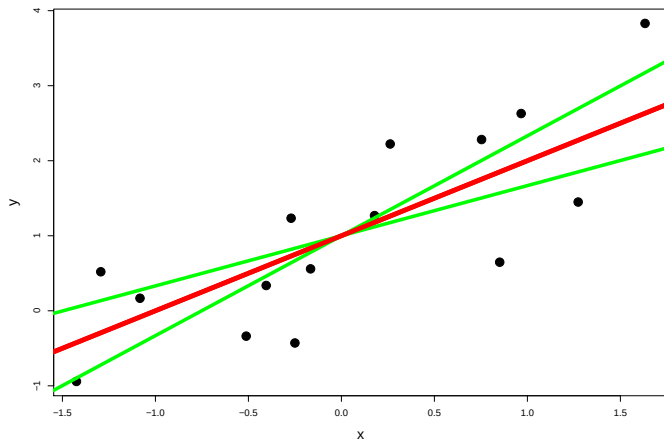
$$P(z_k < k_c) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{k_c} e^{-t^2/2} dt =: \Phi(k_c) = 1 - c$$

- **Upper limit** for w_k :

$$\begin{aligned} P(z_k < k_c) &= P(\sqrt{\sigma^2 S^{kk}} z_k < \sqrt{\sigma^2 S^{kk}} k_c) \\ &= P(w_k - (w_k - \hat{w}_k) > w_k - \sqrt{\sigma^2 S^{kk}} k_c) \\ &= P(\hat{w}_k > w_k - \sqrt{\sigma^2 S^{kk}} k_c) \\ &= P(w_k < \hat{w}_k + \sqrt{\sigma^2 S^{kk}} k_c) = 1 - c. \end{aligned}$$

- Same argument for $z'_k = -z_k \rightsquigarrow$ **lower limit.**

Frequentist confidence limits



Least-squares fit (red) and two lines with slopes according to upper (lower) 95% confidence limit (green).

Standard parametric rate

- Assume we have estimated the parameters based on n samples:

$$\begin{aligned}(\hat{\mathbf{w}}_n - \mathbf{w}) &\sim N(\mathbf{0}, \sigma^2 (X^t X)^{-1}) \\&= N(\mathbf{0}, \sigma^2 (X^t X / n)^{-1} \cdot 1/n) \\ \sqrt{n}(\hat{\mathbf{w}}_n - \mathbf{w}) &\sim N(\mathbf{0}, \sigma^2 \underbrace{(X^t X / n)^{-1}}_{\substack{n \rightarrow \infty \\ \rightarrow \Sigma}})\end{aligned}$$

- Since for $n \rightarrow \infty$, $X^t X / n \rightarrow \Sigma = \text{const.}$, this means that **$\hat{\mathbf{w}}_n$ converges to $\mathbf{w}$ at a rate of $1/\sqrt{n}$.**
- This is a very general result that holds in an asymptotic sense even without assuming normality, due to the **central limit theorem**.
- Due to its universality, it is called the **standard parametric rate**.
- Equivalent statement:
 $1/\sqrt{n}$ represents the **magnitude of the estimation error**.

Basis functions

- Can be generalized to model non-linear relationships by replacing $\mathbf{x}$ with some non-linear function of the inputs, $\phi(\mathbf{x})$:

$$p(y|\mathbf{x}) = N(\mathbf{w}^t \phi(\mathbf{x}), \sigma^2).$$

- Predictions can be based on a linear combination of a set of basis functions $\phi(\mathbf{x}) = \{g_0(\mathbf{x}), g_1(\mathbf{x}), \dots, g_m(\mathbf{x})\}$, with $g_i(\mathbf{x}) : \mathbb{R}^d \mapsto \mathbb{R}$. Can model the intercept by setting $g_0(\mathbf{x}) = 1$:

$$f(\mathbf{x}; \mathbf{w}) = w_0 + w_1 g_1(\mathbf{x}) + \dots + w_m g_m(\mathbf{x}).$$

$\leadsto$ **additive models**

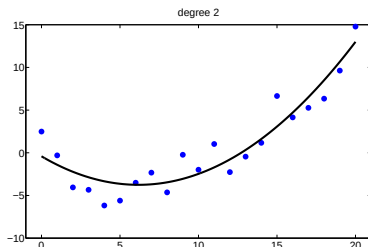
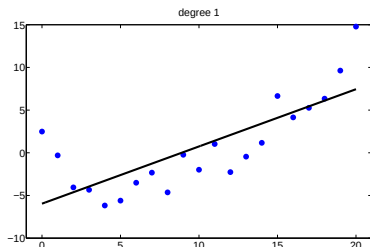


Fig 1.7 in K. Murphy: Machine Learning. MIT Press 2012

Additive models

- Examples:

If $x \in \mathbb{R}^d$ and $m = d + 1$, $g_0(\mathbf{x}) = 1$ and $g_i(\mathbf{x}) = x_i, i = 1, \dots, d$, then

$$f(\mathbf{x}; \mathbf{w}) = w_0 + w_1 x_1 + \dots + w_d x_d.$$

If $x \in \mathbb{R}$, $g_0(\mathbf{x}) = 1$ and $g_i(x) = x^i, i = 1, \dots, m$, then

$$f(x; \mathbf{w}) = w_0 + w_1 x^1 + \dots + w_m x^m.$$

- Basis functions can **capture various properties of the inputs**.

Example: **Document analysis**

$\mathbf{x}$ = text document (collection of words)

$$g_i(\mathbf{x}) = \begin{cases} 1, & \text{if word } i \text{ appears in the document} \\ 0, & \text{otherwise} \end{cases}$$

$$f(\mathbf{x}; \mathbf{w}) = w_0 + \sum_{i \in \text{words}} w_i g_i(\mathbf{x}).$$

Additive models cont'd

- We can also make predictions by gauging the **similarity of examples to prototypes**.
- For example, our additive regression function could be

$$f(\mathbf{x}; \mathbf{w}) = w_0 + w_1 g_1(\mathbf{x}) + \cdots + w_m g_m(\mathbf{x}),$$

where the basis functions are **radial basis functions**

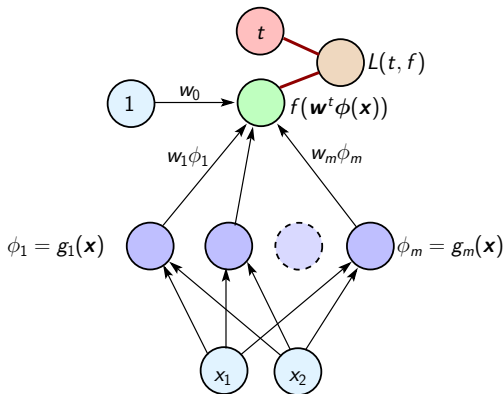
$$g_k(\mathbf{x}) = \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{x} - \mathbf{x}_k\|^2\right)$$

measuring the similarity to the prototypes $\mathbf{x}_k$.

- The variance σ^2 controls how quickly the basis function vanishes as a function of the distance to the prototype.
- **Training examples themselves could serve as prototypes.**

Additive models cont'd

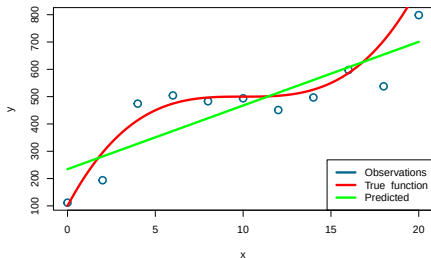
Can view additive models graphically in terms of **units** and **weights**.



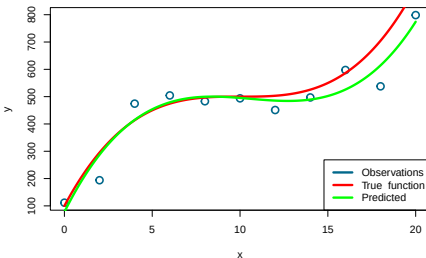
In **Multi Layer Perceptrons** the basis functions have adjustable parameters.

Example: Polynomial regression

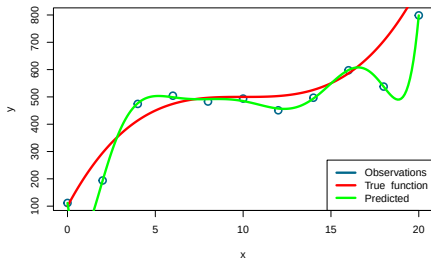
Polynomial basis functions. Degree = 1



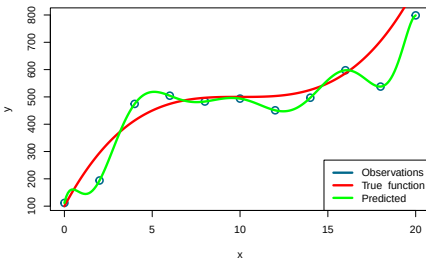
Polynomial basis functions. Degree = 3



Polynomial basis functions. Degree = 8



Polynomial basis functions. Degree = 10



Complexity and overfitting

With limited training examples our polynomial regression model may achieve zero training error but nevertheless has a large expected error.

$$\text{training} \quad \frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i; \hat{\mathbf{w}}))^2 \approx 0$$

$$\text{expectation} \quad E_{(\mathbf{x}, y) \sim p} (y - f(\mathbf{x}; \hat{\mathbf{w}}))^2 \gg 0$$

We suffer from **over-fitting**

$\rightsquigarrow$ should reconsider our model $\rightsquigarrow$ **model selection.**

We will discuss model selection from a **Bayesian perspective** first.

A frequentist approach will follow later in the chapter on **statistical learning theory.**

Bayesian interpretation: priors

- Suppose our generative model takes an input $\mathbf{x} \in \mathbb{R}^d$ and maps it to a real valued output y according to

$$p(y|\mathbf{x}, \mathbf{w}, \sigma^2) = N(y|\mathbf{w}^t \mathbf{x}, \sigma^2)$$

- We will keep σ^2 fixed and only try to estimate $\mathbf{w}$.
- Given data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$, the **likelihood function** is

$$L(\mathbf{w}; \mathcal{D}) = \prod_{i=1}^n N(y_i|\mathbf{w}^t \mathbf{x}_i, \sigma^2) = \prod_{i=1}^n \frac{1}{Z} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mathbf{w}^t \mathbf{x}_i)^2\right).$$

- Predictions in classical regression based on maximizing parameters $\hat{\mathbf{w}}$.
- **In Bayesian analysis we keep all regression functions,**
just weighted by their **posterior probability**:

$$p(y|\mathbf{x}, \mathcal{D}, \sigma^2) = \int p(y|\mathbf{x}, \mathbf{w}, \sigma^2) p(\mathbf{w}|\mathcal{D}, \sigma^2) d\mathbf{w}$$

Bayesian regression: Prior and posterior

- We specify our **prior belief** about the parameter values as $p(\mathbf{w})$. For instance, we could prefer small parameter values:

$$p(\mathbf{w}) = N(\mathbf{w} | 0, \tau^2 I)$$

The smaller τ^2 is, the smaller values of $\mathbf{w}$ we prefer
prior to seeing the data.

- **Posterior** proportional to prior $p(\mathbf{w})$ times likelihood:

$$p(\mathbf{w} | \mathcal{D}, \cdot) \propto L(\mathbf{w}; \mathcal{D}) p(\mathbf{w})$$

- Here: posterior is **Gaussian** $p(\mathbf{w} | \mathcal{D}, \sigma^2) = N(\mathbf{w} | \mathbf{w}_n, V_n)$ with conditional mean $\mathbf{w}_n$ and conditional covariance V_n (i.e. conditioned on dataset of size n) given by

$$\mathbf{w}_n = (X^t X + \lambda I)^{-1} X^t \mathbf{y}, \quad V_n = \sigma^2 (X^t X + \lambda I)^{-1},$$

with $\lambda = \frac{\sigma^2}{\tau^2}$.

Bayesian regression: Posterior computation

Given variables $\mathbf{w} \in \mathbb{R}^d$ and $\mathbf{y} \in \mathbb{R}^n$, assume **linear Gaussian system**:

$$p(\mathbf{w}) = N(\mathbf{w} | \boldsymbol{\mu}_w, \Sigma_w) \quad (\leadsto \text{prior})$$

$$p(\mathbf{y} | \mathbf{w}) = N(\mathbf{y} | A\mathbf{w} + \mathbf{b}, \Sigma_y) \quad (\leadsto \text{likelihood})$$

- **The posterior is also Gaussian** with conditional mean $\boldsymbol{\mu}_{w|y}$ and conditional covariance $\Sigma_{w|y}$:

$$p(\mathbf{w} | \mathbf{y}) = N(\mathbf{w} | \boldsymbol{\mu}_{w|y}, \Sigma_{w|y})$$

$$\Sigma_{w|y}^{-1} = \Sigma_w^{-1} + A^t \Sigma_y^{-1} A$$

$$\boldsymbol{\mu}_{w|y} = \Sigma_{w|y} \left(A^t \Sigma_y^{-1} (\mathbf{y} - \mathbf{b}) + \Sigma_w^{-1} \boldsymbol{\mu}_w \right).$$

Gaussian likelihood and Gaussian prior form a conjugate pair.

- The normalization constant (denominator in Bayes formula) is

$$p(\mathbf{y}) = N(\mathbf{y} | A\boldsymbol{\mu}_w + \mathbf{b}, \Sigma_y + A\Sigma_w A^t).$$

Bayesian regression: Posterior predictive

- Prediction of y for new $\mathbf{x}$: use posterior as weights for predictions based on individual $\mathbf{w}$'s $\rightsquigarrow$ **Posterior predictive:**

$$\begin{aligned} p(y|\mathbf{x}, \mathcal{D}, \sigma^2) &= \int p(y|\mathbf{x}, \mathbf{w}, \sigma^2) p(\mathbf{w}|\mathcal{D}, \sigma^2) d\mathbf{w} \\ &= \int N(y|\mathbf{x}^t \mathbf{w}, \sigma^2) N(\mathbf{w}|\mathbf{w}_n, V_n) d\mathbf{w} \\ &= N(y|\mathbf{w}_n^t \mathbf{x}, \sigma_n^2(\mathbf{x})), \text{ with} \\ \sigma_n^2(\mathbf{x}) &= \sigma^2 + \mathbf{x}^t V_n \mathbf{x}. \end{aligned}$$

- The variance in this prediction, $\sigma_n^2(\mathbf{x})$, depends on two terms:
 - ▶ the variance of the observation noise, σ^2
 - ▶ the variance in the parameters, V_n
 - $\rightsquigarrow$ depends on how close $\mathbf{x}$ is to training data $\mathcal{D}$
 - $\rightsquigarrow$ error bars get larger as we move away from training points.

Bayesian regression: Posterior predictive

- By contrast, the **plugin approximation** uses only the ML-parameter estimate with the degenerate distribution $p(\mathbf{w}|\mathcal{D}, \sigma^2) = \delta_{\hat{\mathbf{w}}}(\mathbf{w})$:
$$p(y|\mathbf{x}, \mathcal{D}, \sigma^2) \approx \int p(y|\mathbf{x}, \mathbf{w}, \sigma^2) \delta_{\hat{\mathbf{w}}}(\mathbf{w}) d\mathbf{w} = p(y|\mathbf{x}, \hat{\mathbf{w}}, \sigma^2) = N(y|\mathbf{x}^t \hat{\mathbf{w}}, \sigma^2).$$

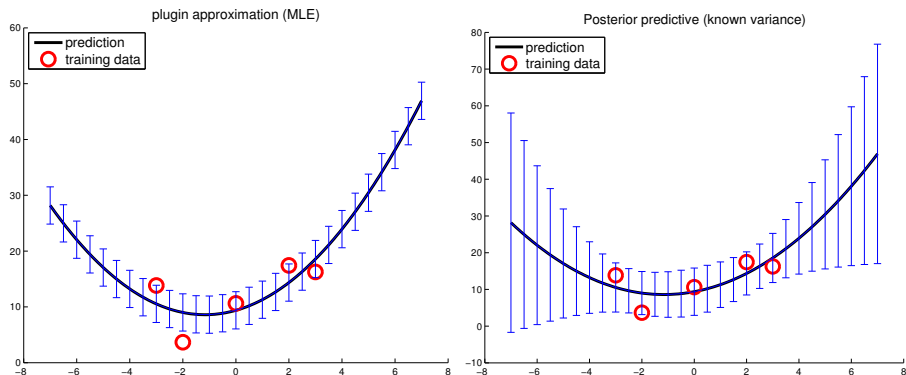


Fig. 7.12 in K. Murphy: Machine Learning. MIT Press 2012. Example with quadratic basis functions: posterior predictive distribution (mean and $\pm 1\sigma$).

Sampling from posterior predictive

Left: plugin approximation: $f(y) = \phi(\mathbf{x})^t \hat{\mathbf{w}}$,
where $\phi(\mathbf{x})$ is the expanded input vector $(1, x, x^2)^t$.

Right: sampled functions $\phi(\mathbf{x})^t \mathbf{w}^{(s)}$, where $w^{(s)}$ are samples from the posterior

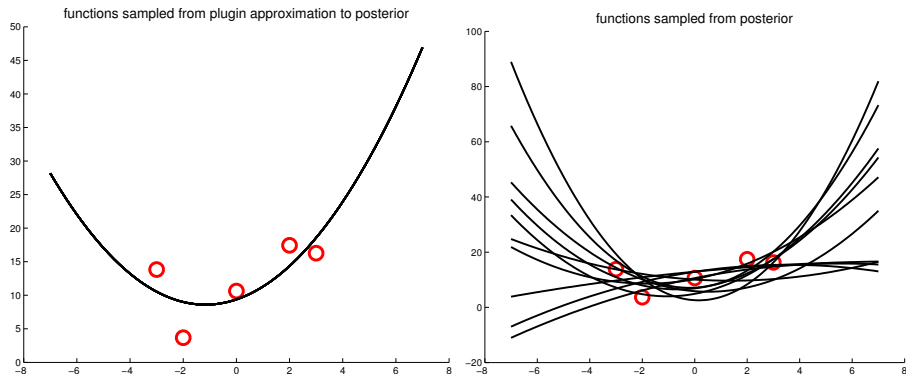


Fig. 7.12 in K. Murphy: Machine Learning. MIT Press 2012

MAP approximation and ridge regression

- Posterior proportional to prior $p(\mathbf{w}) = N(\mathbf{w}|0, \tau^2 I)$ times likelihood.
- The MAP estimate is

$$\begin{aligned}\mathbf{w}_{\text{MAP}} &= \arg \max \{ \log[L(\mathbf{w}; \mathcal{D})] + \log[p(\mathbf{w})] \} \\ &= \arg \min \{ -\log[L(\mathbf{w}; \mathcal{D})] - \log[p(\mathbf{w})] \} \\ &= \arg \min \left\{ \frac{1}{2\sigma^2} \sum_i (y_i - \mathbf{w}^t \mathbf{x}_i)^2 + \frac{1}{2\tau^2} \mathbf{w}^t \mathbf{w} \right\} \\ &= \arg \min \left\{ \sum_i (y_i - \mathbf{w}^t \mathbf{x}_i)^2 + \frac{\sigma^2}{\tau^2} \mathbf{w}^t \mathbf{w} \right\} \\ &= \arg \min \left\{ \sum_i (y_i - \mathbf{w}^t \mathbf{x}_i)^2 + \lambda \mathbf{w}^t \mathbf{w} \right\}\end{aligned}$$

- In classical statistics, this is called **ridge regression**:

$$\mathbf{w}_{\text{MAP}} = \mathbf{w}_{\text{ridge}} = (X^t X + \lambda I)^{-1} X^t \mathbf{y}.$$

- In regularization theory, this is an example of **Tikhonov Regularization**.

Bayesian regression (again)

- Suppose our model within the **model family $\mathcal{F}$** takes an input $\mathbf{x} \in \mathbb{R}^d$ and maps it to a real valued output y according to

$$p(y|\mathbf{x}, \mathbf{w}, \sigma^2) = N(y; \mathbf{w}^t \mathbf{x}, \sigma^2)$$

- We will keep σ^2 fixed and only try to estimate $\mathbf{w}$.
- Given data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$, define likelihood

$$L(\mathbf{w}; \mathcal{D}) = \prod_{i=1}^n N(y_i; \mathbf{w}^t \mathbf{x}_i, \sigma^2) = \prod_{i=1}^n \frac{1}{Z} \exp \left(-\frac{1}{2\sigma^2} (y_i - \mathbf{w}^t \mathbf{x}_i)^2 \right).$$

- Predictions in classical regression based on maximizing parameters $\hat{\mathbf{w}}$.
- **In Bayesian analysis we keep all regression functions,**
just weighted by their **posterior probability:**

$$p(y|\mathbf{x}, \mathcal{D}, \sigma^2) = \int p(y|\mathbf{x}, \mathbf{w}, \sigma^2) p(\mathbf{w}|\mathcal{D}, \sigma^2) d\mathbf{w}$$

Bayesian regression (again)

- We specify our **prior belief** about the parameter values as $p(\mathbf{w}|\mathcal{F})$. For instance, we could prefer small parameter values:

$$p(\mathbf{w}|\mathcal{F}) = N(\mathbf{w}; 0, \tau^2 I)$$

Small $\tau^2 \rightsquigarrow$ small values $\mathbf{w}$ preferred **prior to seeing the data.**

- Posterior proportional to prior times likelihood:

$$p(\mathbf{w}|\mathcal{D}, \cdot) = \frac{p(\mathbf{y}|\mathbf{w}, X)p(\mathbf{w}|\mathcal{F})}{p(\mathbf{y}|\mathcal{F}, X)} \propto L(\mathbf{w}; \mathcal{D})p(\mathbf{w}|\mathcal{F})$$

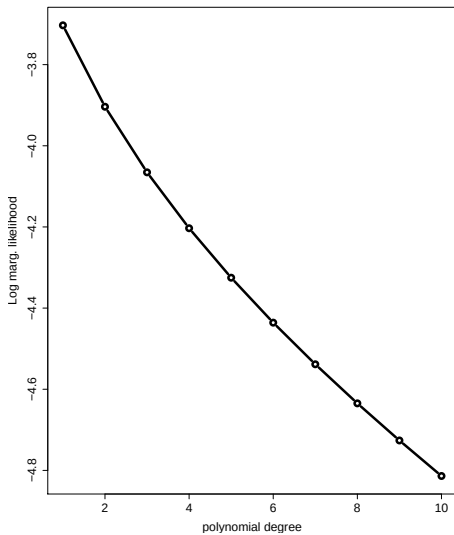
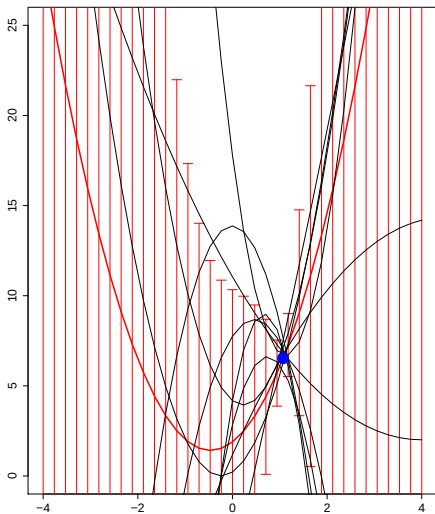
- Normalization constant, a.k.a. **marginal likelihood:**

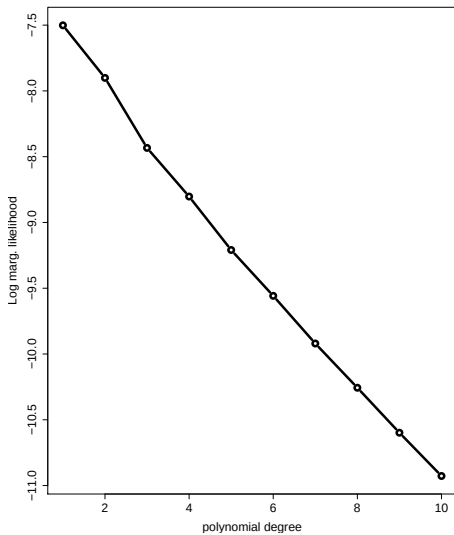
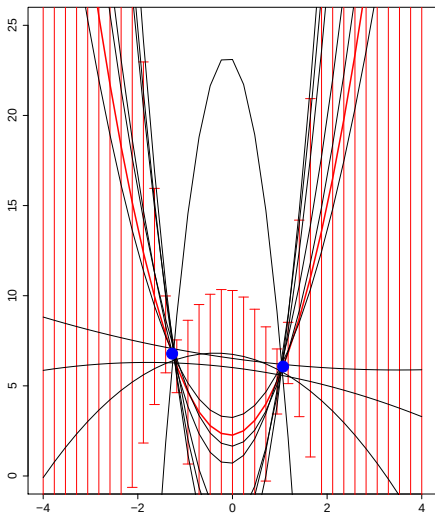
$$p(\mathbf{y}|\mathcal{F}, X) = \int \underbrace{L(\mathbf{w}; \mathcal{D})}_{p(\mathbf{y}|\mathbf{w}, X)} p(\mathbf{w}|\mathcal{F}) d\mathbf{w} = \int p(\mathbf{y}, \mathbf{w}|\mathcal{F}, X) d\mathbf{w},$$

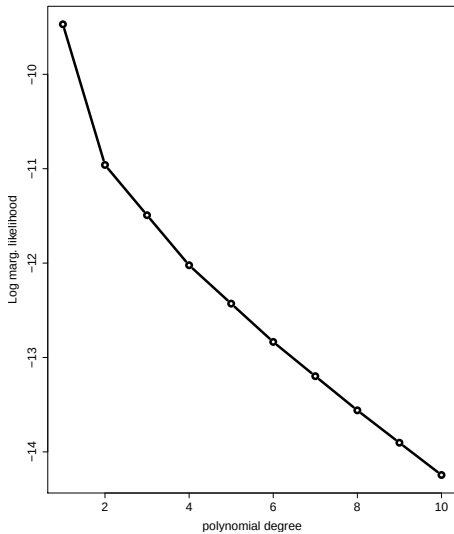
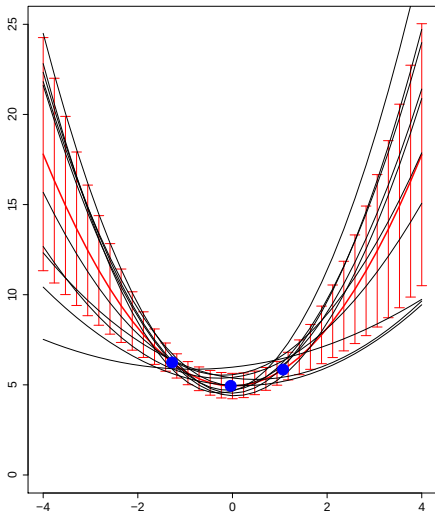
depends on model family $\mathcal{F}$, but not on parameter values of a specific model in the family.

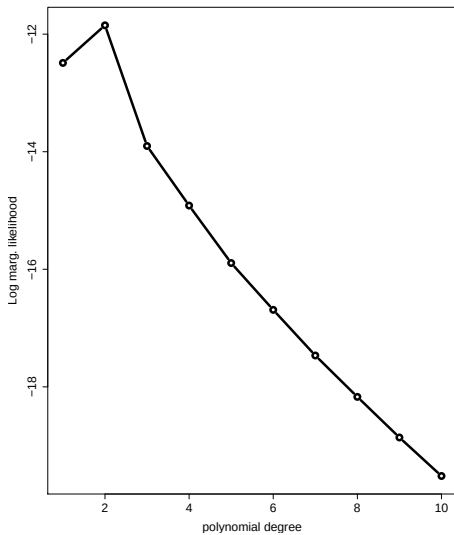
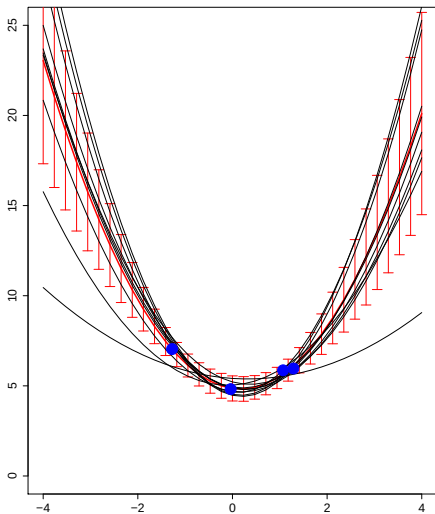
Example: Bayesian regression

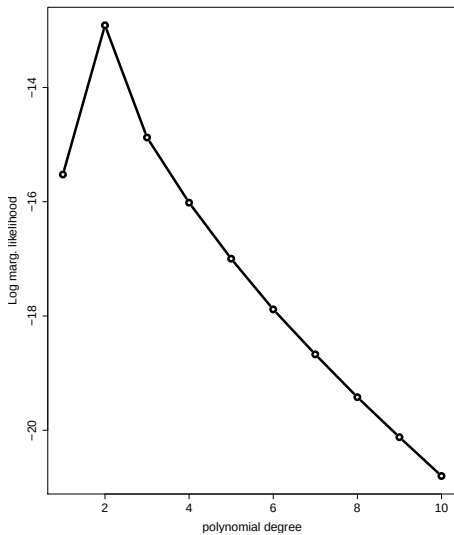
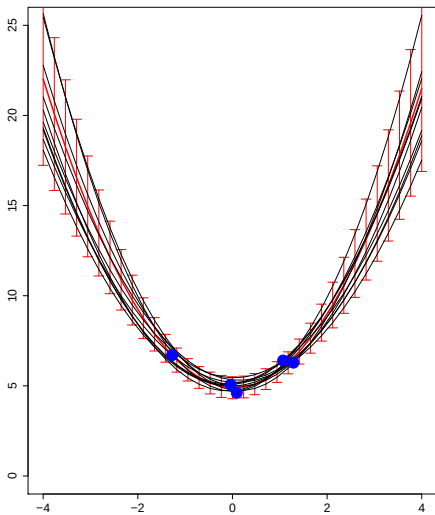
- **Goal: choose among regression model families**, specified by different feature mappings (basis functions) $\mathbf{x} \rightarrow \phi(\mathbf{x})$.
- Example: linear $\phi_1(\mathbf{x}) \in \mathbb{R}^{d_1}$ and quadratic $\phi_2(\mathbf{x}) \in \mathbb{R}^{d_2}$.
- For both families, we specify a Gaussian regression model:
$$\mathcal{F}_i : p(y|\mathbf{x}, \mathbf{w}_i, \sigma^2) = N(y|\mathbf{w}_i^t \phi_i(\mathbf{x}), \sigma^2), \quad i \in \{1, 2\}.$$
- Considering the posterior predictive, there are two possibilities:
 - ▶ **$\mathcal{F}$ too flexible:** posterior requires many training examples before it focuses on useful parameter values;
 - ▶ **$\mathcal{F}$ too simple:** posterior concentrates quickly but the predictions remain poor. **But how can we formalize this intuition?**
- **Posterior of family**, given $\mathcal{D} = \{(y_1, \mathbf{x}_1), \dots, (y_n, \mathbf{x}_n)\} = \{\mathbf{y}, X\}$:
$$p(\mathcal{F}|\mathbf{y}, X) \propto p(\mathbf{y}|\mathcal{F}, X) P(\mathcal{F}).$$
- Pragmatic choice: Uniform prior over families $\rightsquigarrow$ select the family whose **marginal likelihood** (a.k.a. Bayesian score) **is larger**, i.e. select family $\mathcal{F}_1$ if $p(\mathbf{y}|\mathcal{F}_1, X) > p(\mathbf{y}|\mathcal{F}_2, X)$.

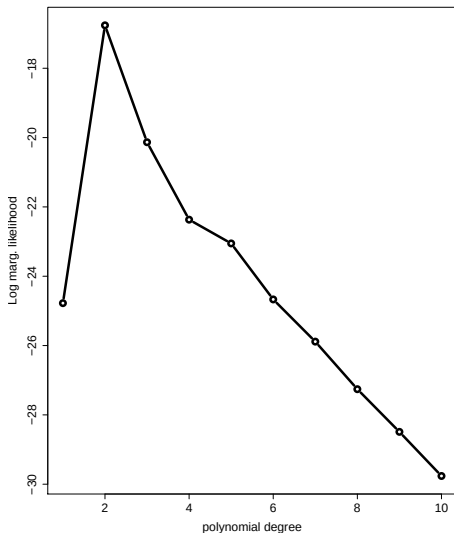
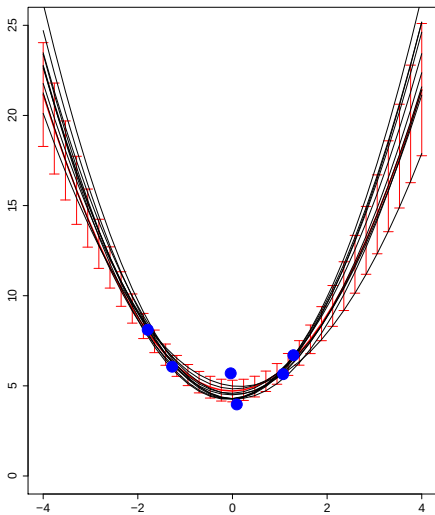


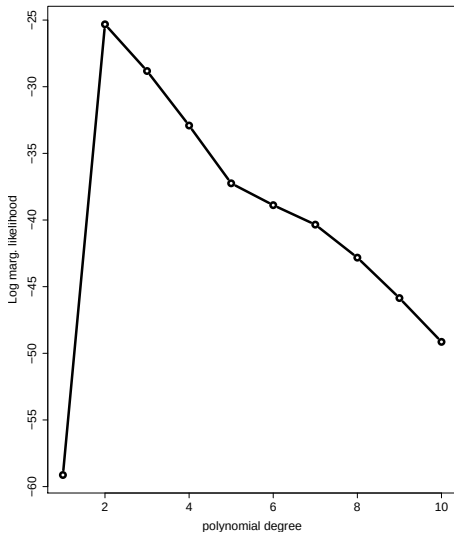
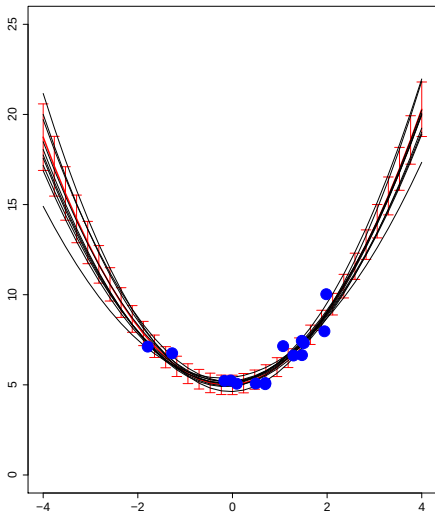












Approximating the marginal likelihood

- Problem: In most cases we cannot compute the marginal likelihood in closed form $\rightsquigarrow$ approximations are needed.
- A specific approximation will lead to the **Bayesian Information Criterion (BIC)**.
- Key insight: when computing

$$p(\mathbf{y}|\mathcal{F}, X) = \int L(\mathbf{w}; \mathcal{D}) p(\mathbf{w}|\mathcal{F}) d\mathbf{w},$$

the integrand is a product of two densities $\rightsquigarrow$ integrand itself is an unnormalized density.

- Laplace's approximation uses a clever trick to approximate such integrals...

Approximation details: Laplace's Method

- Assume unnormalized density $p^*(\theta)$ has peak at $\hat{\theta}$. Goal: calculate normalizing constant

$$Z_p = \int p^*(\theta) d\theta$$

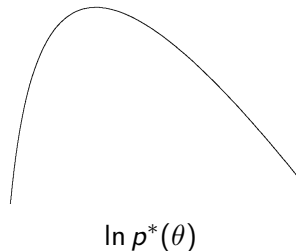
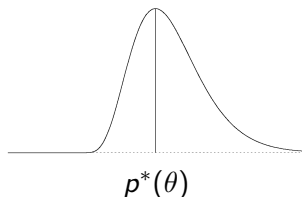
- Taylor-expand logarithm around $\hat{\theta}$:

$$\ln p^*(\theta) \approx \ln p^*(\hat{\theta}) - \frac{c}{2}(\theta - \hat{\theta})^2 + \dots,$$

where

$$c := -\frac{\partial^2}{\partial \theta^2} \ln p^*(\theta) \Big|_{\theta=\hat{\theta}}.$$

(note that first order term vanishes)



Laplace's Method (cont'd)

- Approximate $p^*(\theta)$ by unnormalized Gaussian

$$Q^*(\theta) := p^*(\hat{\theta}) \exp \left[-c/2 \cdot (\theta - \hat{\theta})^2 \right]$$

- A normalized Gaussian would be:

$$Q(\theta \mid \mu = \hat{\theta}, \sigma^2) = \frac{1}{Z_Q} \exp \left[-\frac{(\theta - \hat{\theta})^2}{2\sigma^2} \right],$$

with $Z_Q = \sqrt{2\pi\sigma^2}$

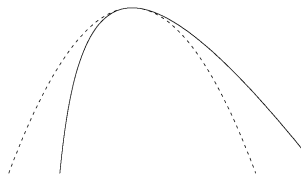
$$= \int \exp \left[-1/(2\sigma^2) \cdot (\theta - \hat{\theta})^2 \right] d\theta$$

- Approximate $Z_p = \int p^*(\theta) d\theta$ by

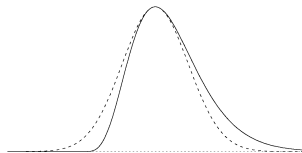
$$Z_p \approx \int Q^*(\theta) d\theta$$

$$= p^*(\hat{\theta}) \int \exp \left[-c/2 \cdot (\theta - \hat{\theta})^2 \right] d\theta$$

$$= p^*(\hat{\theta}) \sqrt{2\pi/c} \rightsquigarrow c \text{ is the inverse variance}$$



$\ln p^*(\theta)$ & $\ln Q^*(\theta)$



$p^*(\theta)$ & $Q^*(\theta)$

Laplace's Method (cont'd)

- Multivariate generalization in d dimensions:
second derivative $\rightsquigarrow$ **Hessian matrix**

$$H_{ij} = \left. \frac{\partial^2 \ln p^*(\theta)}{\partial \theta_i \partial \theta_j} \right|_{\theta = \hat{\theta}}$$

$$\begin{aligned} Z_p &\approx p^*(\hat{\theta}) \int \exp \left[-\frac{1}{2} (\theta - \hat{\theta})^t H (\theta - \hat{\theta}) \right] d\theta \\ &= p^*(\hat{\theta}) \sqrt{\frac{(2\pi)^d}{|H|}} = p^*(\hat{\theta}) \left| \frac{H}{2\pi} \right|^{-\frac{1}{2}}, \end{aligned}$$

where the last equation follows from the properties of the determinant: $|aM| = a^d |M|$ for $M \in \mathbb{R}^{d \times d}$, $a \in \mathbb{R}$.

- Interpretation:

$p(\theta)$ is approximated by a Gaussian centered at the mode $\hat{\theta}$:

$$p(\theta) \approx \mathcal{N}(\theta | \mu = \hat{\theta}, \Sigma = H^{-1}).$$

Bayesian Information Criterion (BIC)

$$p(\mathcal{D}|\mathcal{F}) = \int p(\mathcal{D}|\mathbf{w}) \cdot p(\mathbf{w}|\mathcal{F}) d\mathbf{w}$$
$$\approx p(\mathcal{D}|\mathbf{w}^*) \cdot p(\mathbf{w}^*|\mathcal{F}) |H/(2\pi)|^{-\frac{1}{2}} \stackrel{\text{flat prior}}{\approx} p(\mathcal{D}|\hat{\mathbf{w}}) |H/(2\pi)|^{-\frac{1}{2}}$$

$$\log p(\mathcal{D}|\mathcal{F}) \approx \log p(\mathcal{D}|\hat{\mathbf{w}}) - \frac{1}{2} \log |H| + C, \quad \text{with } \hat{\mathbf{w}} = \mathbf{w}_{MLE} \text{ in } \mathcal{F}.$$

- i.i.d. $\rightsquigarrow \log p(\mathcal{D}|\hat{\mathbf{w}}) = \sum_{i=1}^n \log p(\mathcal{D}_i|\hat{\mathbf{w}})$, but C independent of n .

Idea: for large n , drop all terms that do not depend on n .

- Focus on Hessian:

$$H = \sum_{i=1}^n H_i, \quad \text{with } H_{i_{jk}} = \left. \frac{\partial^2 \log p(\mathcal{D}_i|\mathbf{w})}{\partial w_j \partial w_k} \right|_{\mathbf{w}=\hat{\mathbf{w}}}$$

Let's approximate each H_i with a **fixed** matrix H'

$$\log |H| = \log |nH'| = \log(n^d |H'|) = d \log n + \log(|H'|).$$

- Again, drop terms that don't depend on n :

$$\log p(\mathcal{D}|\mathcal{F}) + C' \approx \log p(\mathcal{D}|\hat{\mathbf{w}}) - \frac{d}{2} \log n =: \text{BIC}(\mathcal{F}, n|\mathcal{D}).$$

Intuitive interpretation of BIC

- The **Shannon information content** of a specific outcome a of a random experiment is

$$h(a) = -\log_2 P(a) = \log \frac{1}{P(a)}.$$

It measures the “surprise” (in bits):

Outcomes that are less probable have larger values of surprise.

- **Information theory:** Can find a code so that the **number of bits** used to encode each symbol $a \in \mathcal{A}$ is essentially $-\log_2 P(a)$.
- Here:

$$-\text{BIC}(\mathcal{F}, n|\mathcal{D}) = \overbrace{\sum_{i=1}^n \left(\underbrace{-\log_2 p(y_i | \mathbf{x}_i, \hat{\mathbf{w}})}_{\text{surprise of } y_i} \right)}^{\text{DL of observations given model}} + \frac{d}{2} \log_2(n)$$

- The sum of surprises of all observations is the **description length of the observations given the (most probable) model in $\mathcal{F}$.**

Intuitive interpretation of BIC

Second term: **DL of the model.** Intuitive explanation:

- The model, i.e. $\hat{\mathbf{w}} \in \mathbb{R}^d$, was estimated based on n samples.
- Can quantize every component into $\sqrt{n}$ levels. Why?
- Recall the **standard parametric rate:**

$$\sqrt{n}(\hat{\mathbf{w}}_n - \mathbf{w}) \sim N(\mathbf{0}, \sigma^2 \underbrace{(X^t X / n)^{-1}}_{\substack{n \rightarrow \infty \\ \rightarrow \Sigma}})$$

$\leadsto \hat{\mathbf{w}}_n$ converges to true $\mathbf{w}$ at a rate of $1/\sqrt{n}$

$\leadsto 1/\sqrt{n}$ represents the magnitude of the estimation error

$\leadsto$ **no need for encoding with greater precision:**

- ▶ Assume $w \in \mathbb{R}$, and range of w rescaled to unit interval $[0, 1]$.
- ▶ Instead of communicating exact numerical value of $\hat{w}_n$ over the communication channel, we can partition the unit interval into $\sqrt{n}$ bins and **communicate only the number of the bin.**

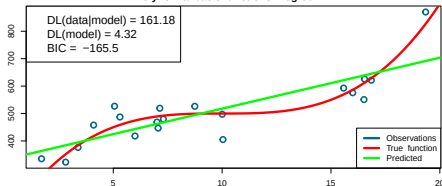
Intuitive interpretation of BIC

- In $\mathbb{R}^d$: Grid of $(\sqrt{n})^d$ possible values for describing the model.
- We only need $\log_2((\sqrt{n})^d) = \log_2 n^{(d/2)} = (d/2) \log_2 n$ **bits to encode $\hat{\mathbf{w}}$ with sufficient precision.**
- In summary:

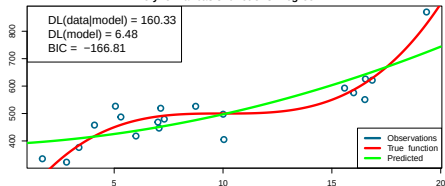
$$\begin{aligned} -\text{BIC} &= -\log_2 p(\mathcal{D}|\hat{\mathbf{w}}) + \frac{d}{2} \log_2 n \\ &= \text{DL}(\text{data}|\text{model}) + \text{DL}(\text{model}). \end{aligned}$$

- Maximizing BIC = minimizing the joint DL of data and model
 $\leadsto$ **Minimum Description Length principle.**

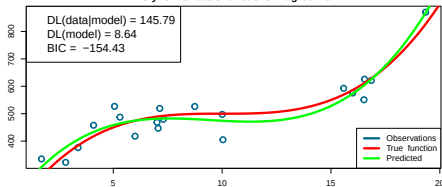
Polynomial basis functions. Degree = 1



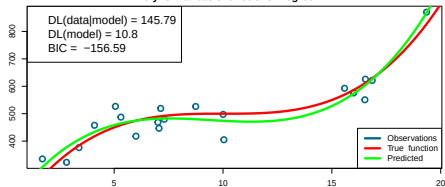
Polynomial basis functions. Degree = 2



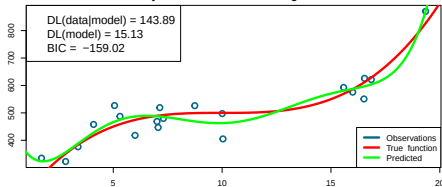
Polynomial basis functions. Degree = 3



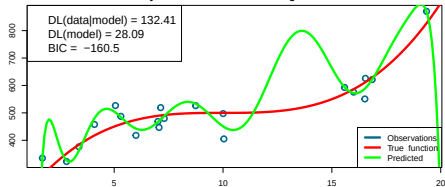
Polynomial basis functions. Degree = 4



Polynomial basis functions. Degree = 6



Polynomial basis functions. Degree = 12



Sparse Models

- Sometimes, we have many more dimensions d than training cases n .
- Corresponding design matrix X is “short and fat”, rather than “tall and skinny”.
- This is called **small n , large d problem**.
- For example, with **gene microarrays**, it is common to measure the expression levels of $d \approx 20,000$ genes, but to only get $n \approx 100$ samples (for instance, from 100 patients).
- Q: what is the **smallest set of features** that can accurately predict the response in order to **prevent overfitting**, to **reduce the cost** of building a diagnostic device, or to help with **scientific insight** into the problem?

Bayesian variable selection

- Let $\gamma_j = 1$ if feature j is **relevant**, and let $\gamma_j = 0$ otherwise.

- Our goal is to compute the posterior over models

$$p(\gamma|\mathcal{D}) \propto p(\mathcal{D}|\gamma)p(\gamma)$$

- Example: generate $n = 20$ samples from a $d = 10$ dimensional linear model, $y_i \sim N(w^t x_i, \sigma^2)$, in which $K = 5$ elements of w are non-zero.
- Enumerate all $2^{10} = 1024$ models. Note that a model is expressed as a specific **sparsity pattern** via a bit string, such as

$$(0, 1, 1, 0, 0, 1, 0, 1, 1, 0).$$

Then, compute $p(\gamma|\mathcal{D})$ for each one.

- Interpreting the posterior over a large number of models is difficult
 $\rightsquigarrow$ seek **summary statistics**.
- Natural choice: **MAP estimate**: $\hat{\gamma} = \arg \max_{\gamma} p(\gamma|\mathcal{D})$.

Bayesian variable selection

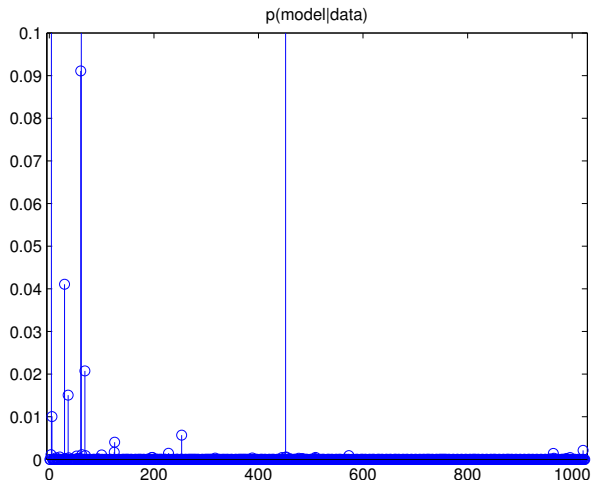


Fig 13.1 in K. Murphy: Machine Learning. MIT Press 2012. Posterior over all 1024 models. Vertical scale has been truncated at 0.1 for clarity.

Bayesian variable selection

- The above example illustrates the **gold standard** for variable selection: the problem was small ($d = 10$)
 $\rightsquigarrow$ we were able to **compute the full posterior exactly**.
- Of course, variable selection is most useful in the cases where the number of dimensions is **large**.
- There are 2^d possible models (bit vectors)
 $\rightsquigarrow$ **impossible** to compute the full posterior in general.
- Even finding summaries is **intractable**
 $\rightsquigarrow$ **algorithmic speedups** necessary.
- But first, focus on the computation of the posterior $p(\gamma|\mathcal{D})$.

The spike and slab model

- The posterior is given by

$$p(\gamma|\mathcal{D}) = p(\gamma|\mathbf{y}, X) \propto p(\gamma)p(\mathbf{y}|\gamma, X)$$

- It is common to use the following prior:

$$p(\gamma) = \prod_{j=1}^d \text{Ber}(\gamma_j|\pi_0) = \pi_0^{\|\gamma\|_0} (1 - \pi_0)^{d - \|\gamma\|_0},$$

$$\log p(\gamma|\pi_0) = -\lambda \|\gamma\|_0 + \text{const.},$$

where π_0 is the probability that a feature is relevant,

and $\|\gamma\|_0 = \sum_{j=1}^d \gamma_j$ is the ℓ_0 pseudo-norm,

i.e., the **number of non-zero elements**.

- $\lambda = \log \frac{1-\pi_0}{\pi_0}$ controls the **sparsity** of the model.
- Setting $\sigma^2 = 1$, we can write the marginal likelihood as follows:

$$p(\mathbf{y}|X, \gamma) = \int p(\mathbf{y}|X, \mathbf{w}, \gamma) p(\mathbf{w}|\gamma) d\mathbf{w}$$

The spike and slab model

- Focus on prior over weights $p(\mathbf{w}|\gamma)$. If $\gamma_j = 0$, feature j is **irrelevant**, so we expect $w_j = 0$. If $\gamma_j = 1$, we expect w_j to be non-zero.
- Assume a **Gaussian prior**, $N(0, \sigma_w^2)$, where σ_w^2 reflects our expectation of the coefficients associated with the **relevant variables**:

$$p(w_j|\gamma_j) = \begin{cases} \delta_0(w_j) & , \text{ if } \gamma_j = 0 \\ N(w_j|0, \sigma_w^2) & , \text{ else} \end{cases}$$

- **The first term is a spike at the origin.**
- As $\sigma_w^2 \rightarrow \infty$, the distribution $p(w_j|\gamma_j = 1)$ approaches a **uniform distribution** $\rightsquigarrow$ **second term is slab of constant height.**
- **Spike and slab model** (Mitchell and Beauchamp 1988).
- Full Bayesian treatment is **computationally challenging!**

Simplifying the model

- Assume $\sigma_w^2 \rightarrow \infty$ ($\rightsquigarrow$ uniform prior $p(w_j|\gamma_j)$ over nonzero components) and approximate the marg. likelihood using **BIC**:

$$\begin{aligned}\log p(\mathbf{y}|\gamma, X) &= \int p(\mathbf{y}|X, \mathbf{w}, \gamma) p(\mathbf{w}|\gamma) d\mathbf{w} \\ &\approx \log p(\mathbf{y}|X, \hat{\mathbf{w}}_\gamma) - \frac{1}{2} \underbrace{\|\hat{\mathbf{w}}_\gamma\|_0}_{\text{"effective" dimension}} \log n,\end{aligned}$$

where $\hat{\mathbf{w}}_\gamma$ is the ML estimate.

- Another view of this model: select $\hat{\mathbf{w}}$ by minimizing the negative log likelihood **under a ℓ_0 -penalty**:

$$\text{minimize } -\log p(\mathbf{y}|X, \mathbf{w}) + \lambda \|\mathbf{w}\|_0.$$

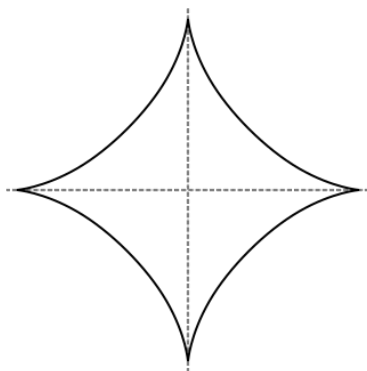
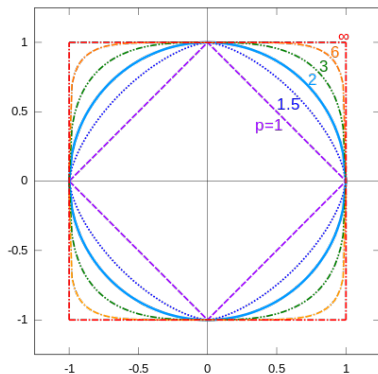
- Practical problem: ℓ_0 pseudo-norm is highly non-convex!**

Vector norms

The **vector p -norms** (ℓ_p norms) are defined by

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \quad 1 \leq p \leq \infty,$$

$$\|\mathbf{x}\|_\infty = \max(|x_1|, \dots, |x_n|).$$



Simplifying the model further

- ℓ_0 penalty $\rightsquigarrow$ **combinatorial optimization problem.**
- When we have many variables, it is computationally difficult to find the the minimizer of $-\log p(\mathbf{y}|X, \mathbf{w}) + \lambda \|\mathbf{w}\|_0$.
- Idea: replace discrete variables with continuous ones. Use continuous priors that “encourage” $w_j = 0$ by putting a lot of probability density near the origin, such as a **zero-mean Laplace distribution.**

$$p(\mathbf{w}|\lambda) = \prod_{j=1}^d \text{Lap}(w_j|0, 1/\lambda) \propto \prod_{j=1}^d \exp(-\lambda|w_j|)$$

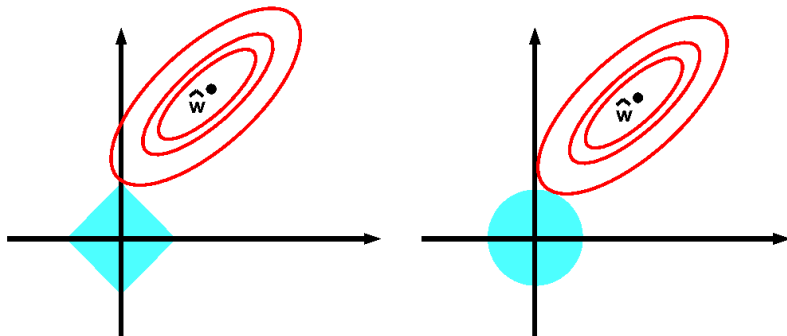
- Let us perform **MAP estimation with this prior:**

$$f(\mathbf{w}) = -\log p(\mathbf{y}|\mathbf{w}, X) - \log p(\mathbf{w}|\lambda) = NLL(\mathbf{w}) + \lambda \|\mathbf{w}\|_1,$$

where $\|\mathbf{w}\|_1 = \sum_{j=1}^d |w_j|$ is the ℓ_1 norm of $\mathbf{w}$,
and NLL means **negative log-likelihood.**

The Lasso

- Can be thought of as a **convex approximation** to the ℓ_0 norm.
- For suitably large λ , the estimate $\hat{\mathbf{w}}$ will still be **sparse**.
- This model has the colorful name **least absolute shrinkage and selection operator**.



The Lasso

- Unfortunately, the $\|\mathbf{w}\|_1$ term is not differentiable at 0
 $\rightsquigarrow$ **convex, but non-smooth optimization problem.**
- The **subderivative or subgradient** of a (convex) function $f : \mathcal{I} \rightarrow \mathbb{R}$ at a point x_0 is a scalar c such that

$$f(x) - f(x_0) \geq c(x - x_0), \quad \forall x \in \mathcal{I}$$

where $\mathcal{I}$ is some interval containing x_0 .

Note that c is a **linear lower bound** to f at x_0 .

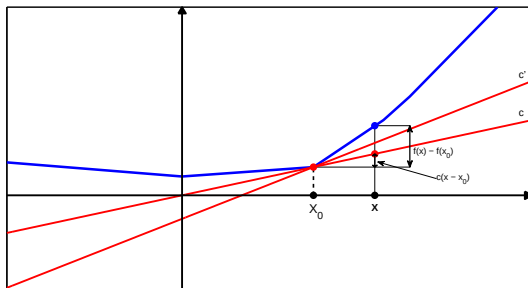


Fig. 13.4 in K. Murphy: Machine Learning. MIT Press 2012

The Lasso

- The set of all subderivatives is called the **subdifferential**
- For the **absolute value function** $f(x) = |x|$:

$$\partial f(x) = \begin{cases} -1 & , \text{ if } x < 0 \\ [-1, 1] & , \text{ if } x = 0 \\ +1 & , \text{ if } x > 0 \end{cases}$$

- For least-squares regression, it is easy to show that

$$\frac{\partial}{\partial w_j} \text{RSS}(\mathbf{w}) = a_j w_j - c_j$$

$$a_j = 2 \sum_{i=1}^n x_{ij}^2$$

$$c_j = 2 \sum_{i=1}^n x_{ij} (y_i - \mathbf{w}_{-j}^t \mathbf{x}_{i,-j}),$$

where $\mathbf{w}_{-j}$ is $\mathbf{w}$ without component j .

The Lasso

- c_j is (proportional to) the correlation between the j 'th feature $\mathbf{x}_{\cdot j} = (x_{1j}, x_{2j}, \dots, x_{nj})^t$ and the **residual due to other features**:

$$c_j \propto \mathbf{x}_{\cdot j}^t \mathbf{r}_{(-j)}, \text{ with } \mathbf{r}_{(-j)} = \mathbf{y} - \underbrace{\mathbf{X}_{-j}}_{\text{X w/o } j\text{-th col}} \mathbf{w}_{-j}.$$

- Recall that the **residual from the least squares estimate is orthogonal to every input feature**.

↪ **magnitude of c_j indicates how relevant feature j is, relative to all other features.**

- Adding the ℓ_1 penalty term:

$$\begin{aligned} \partial_{w_j} f(\mathbf{w}) &= (a_j w_j - c_j) + \lambda \partial_{w_j} \|\mathbf{w}\|_1 \\ &= \begin{cases} a_j w_j - c_j - \lambda & , \text{ if } w_j < 0 \\ [-c_j - \lambda, -c_j + \lambda] & , \text{ if } w_j = 0 \\ a_j w_j - c_j + \lambda & , \text{ if } w_j > 0 \end{cases} \end{aligned}$$

The Lasso

- Depending on the value of c_j , the solution to $\partial_{w_j} f(\mathbf{w}) = 0$ can occur at three different values of w_j :

$$\hat{w}_j = \begin{cases} (c_j + \lambda)/a_j & , \text{ if } c_j < -\lambda \\ 0 & , \text{ if } c_j \in [-\lambda, \lambda] \\ (c_j - \lambda)/a_j & , \text{ if } c_j > \lambda \end{cases}$$

- We can write this as follows:

$$\hat{w}_j = \text{soft} \left(\frac{c_j}{a_j}; \frac{\lambda}{a_j} \right),$$

where $\text{soft}(a; \delta) = \text{sign}(a)(|a| - \delta)_+$

and $x_+ = \max(x, 0)$ is the positive part of x .

- This is called **soft thresholding**.

The Lasso

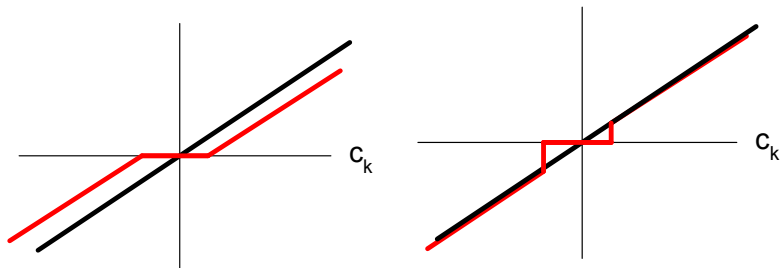


Fig. 13.5 in K. Murphy: Machine Learning. MIT Press 2012.

Black line: **Least squares fit** $w_k = c_k/a_k$.

The red line (the regularized estimate) $\hat{w}_k(c_k)$, shifts the black line down (or up) by λ , except when $-\lambda \leq c_k \leq \lambda$, in which case it sets $w_k = 0$.

By contrast, **hard thresholding** sets values of w_k to 0 if $-\lambda \leq c_k \leq \lambda$, but it **does not shrink the values of w_k outside of this interval**.

Lasso Algorithms: Coordinate-wise Descent

Sometimes it is hard to optimize all variables simultaneously, but it is easy to **optimize them one by one**. Assume that we can efficiently solve for the j -th coefficient w_j with all other coefficients held fixed:

$\hat{w}_j = \arg \min_z f(\mathbf{w} + z\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j -th unit vector.

Then cycle through these component-wise updates.

For the Lasso, this is particularly simple:

for $j = 1, \dots, d$ do:

$$a_j = 2 \sum_{i=1}^n x_{ij}^2$$

$$c_j = 2 \sum_{i=1}^n x_{ij}(y_i - \mathbf{w}_{-j}^t \mathbf{x}_{i,-j})$$

$$\hat{w}_j = \text{soft} \left(\frac{c_j}{a_j}; \frac{\lambda}{a_j} \right).$$